

π^0 A Vision-Language-Action Flow Model for General Robot Control

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Abstract

- Robot learning holds tremendous promise to unlock the full potential of flexible, general, and dexterous robot systems, as well as to address some of the deepest questions in artificial intelligence.
- However, bringing robot learning to the level of generality required for effective real-world systems faces major obstacles in terms of **data**, **generalization**, and **robustness**.
- In this paper, we discuss how generalist robot policies (i.e., robot foundation models) can address these challenges, and how we can design effective generalist robot policies for complex and highly dexterous tasks.
- We propose **a novel flow matching architecture built on top of a pre-trained vision-language model (VLM) to inherit Internet-scale semantic knowledge**.
- We then discuss how this model can be trained on a large and diverse dataset from multiple dexterous robot platforms, including single-arm robots, dual-arm robots, and mobile manipulators.
- We evaluate our model in terms of its **ability to perform tasks via direct prompting, follow language instructions from people and from a high-level VLM policy**, and **its ability to acquire new skills via fine-tuning**.

Introduction

Versatility for AI

- Human intelligence's primary advantage over machines is versatility: ***the ability to solve diverse tasks across various physical environments*** and ***respond intelligently to constraints, commands, and unexpected disturbances***.
- Progress toward this versatility is ***most evident in large-scale language and vision-language models*** [1, 48].
- These systems are ***pre-trained on vast, diverse web-scraped image and text data***, then ***fine-tuned*** with curated datasets for ***desired behaviors and responsiveness***.

Introduction

Why Versatility in Physical AI?

- Like humans, they are ***not physically present in the real world, and their understanding of physical interaction is based entirely on abstract descriptions.***
- ***If these methods are to make practical progress toward AI systems that exhibit the same physical-context-based versatility that humans possess, the models must be trained using data based on a physical context—that is, data from real robot agents.***
- A flexible and general model that can be commanded to perform a wide variety of robot behaviors not only has enormous practical implications but may also offer a solution to some of the hardest challenges facing robot learning today, such as data availability, generalization, and robustness.

If it is possible,

- ***Even for effective specialized robot systems, it may be more effective to first pre-train on a highly diverse set of robot data and then fine-tune for the desired task or receive specific task inputs.*** This way, ***the problem of data scarcity can be solved.***
- Furthermore, diverse data presents a broader range of observations and behaviors, thereby offering various scenes, correction, and recovery actions that may not be present in a narrow range of specialized data, which ***helps address the problems of robustness and generalization.***

Introduction

Challenges for Versatility in Physical AI

However, there are several major challenges in developing these generalized robot policies, or robot foundation models.

1. **First, this research must be conducted at a very large scale because the full benefits of large-scale pre-training often do not manifest at smaller scales [54].**
2. **Second, an appropriate model architecture must be developed that can effectively utilize diverse data sources while also being able to represent the subtle and complex behaviors required to interact with complex physical scenes.**
3. **Third, an appropriate training recipe is needed.**

Introduction

First, this research must be conducted at a very large scale because the full benefits of large-scale pre-training often do not manifest at smaller scales [54].

- By building the model on a **VLM**, it inherits the **general knowledge, semantic reasoning, and problem-solving capabilities** of language models and vision-language models.
- To utilize diverse robot data sources, **cross-embodiment learning [10]** is used, which combines data from **multiple robot types into the same model**.

Introduction

Second, an appropriate model architecture must be developed that can effectively utilize diverse data sources while also being able to represent the subtle and complex behaviors required to interact with complex physical scenes.

- To enable the performance of highly dexterous and complex physical tasks, ***we use an action chunking architecture [57] that employs flow matching (a variant of diffusion) to represent complex continuous action distributions [28, 32].***
- This allows the model to control the robot at frequencies up to 50 Hz for highly dexterous tasks like laundry folding.
- To combine flow matching with the VLM, we use a new action expert that adds a flow-based output to a standard VLM.

Introduction

Third, an appropriate training recipe is needed.

Our training approach follows the ***pre-training/post-training separation paradigm*** commonly seen in exascale language and image-language models [1, 48].

- In this approach, the model is first ***pre-trained on a very large and diverse corpus, and then fine-tuned with more restricted and carefully selected data to induce desired behavioral patterns (in our case, dexterity, efficiency, and robustness)***.

Intuitively, ***training only on high-quality data means that errors rarely occur, so the model does not learn how to recover from them. Conversely, training only on low-quality pre-training data means the model does not learn to operate efficiently and robustly.***

- Combining the two means the ***model attempts to behave as much like the high-quality data as possible, but has a diverse set of recovery and correction methods it can use in case an error occurs.***

Introduction

Contributions

- The main contributions of this study are a ***novel general-purpose robot policy architecture based on VLM pre-training and flow matching*** and ***empirical research on the pre-training/post-training approach for such robot foundation models***.
- The model was evaluated through its ***basic execution performance using language commands, fine-tuning for sub-tasks***, and a ***combination with a high-level semantic policy that outputs intermediate language commands for complex, long-duration tasks***.

Related Works

Autoregressive Discretization vs. Flow Matching

- It is closely related to recently proposed VLA (Vision Language Action) models that use pre-trained VLM (Vision Language Model) optimized for robot control [7, 24, 55].
- **These models use autoregressive discretization to represent actions in a manner analogous to text tokens.**
- In contrast, the model proposed in this study adopts a novel design that **fine-tunes the VLM to generate actions via flow matching [32, 28], a variant of diffusion [20, 46].**
- It can handle **high-frequency action chunks [57] up to 50 Hz and highly dexterous tasks**, which has been a major challenge for prior autoregressive VLA models [7].
- This is similar to recent studies on diffusion models for action generation [9, 60].

Related Works

VLM backbone & Pre/Post Training

- **The model proposed in this study uses a pre-trained VLM backbone [5].**
- The contribution of this study is fundamentally integrative, focusing on a framework for robot foundation models, including not only the **model architecture** itself, but also the **pre-training method**, pre-training and **post-training phases**, and various **real-world experiments**.

Related Works

Hybrid Model (Diffusion + ARL)

- In addition to the field of robot control, **many models combining pre-trained language models and diffusion have been proposed** [40, 41, 14],
- In particular, there are **hybrid models combining diffusion with Autoregressive Large Language Models (ARL)** [19, 29, 59].
- These models generally focus on image generation, but the action generation model in this study is based on several previously proposed concepts.
- Similar to Zhou et al.[59], in this study, we train the model by **applying a diffusion-style (flow matching) loss to individual sequence elements, instead of the standard cross-entropy loss** used in decoder-only transformers.
- Also, similar to Liu et al.[29], **we use a separate set of weights for tokens corresponding to diffusion**.
- In this study, we present the **first flow matching VLA that integrates these concepts into a VLA model to generate high-frequency action chunks for dexterous control**.

Related Works

First Trail in the World

- Since one of the goals of this study is to investigate complex and dexterous behaviors, we utilized a much larger dataset, supplementing the open-source OXE dataset [10] to include approximately 10,000 hours of demonstration data.
- To our knowledge, this is the largest robot learning experiment conducted to date in terms of the amount of robot data.
- To our knowledge, this study implements the longest-duration dexterous manipulation tasks in the field of robot learning.

Overview

Dataset

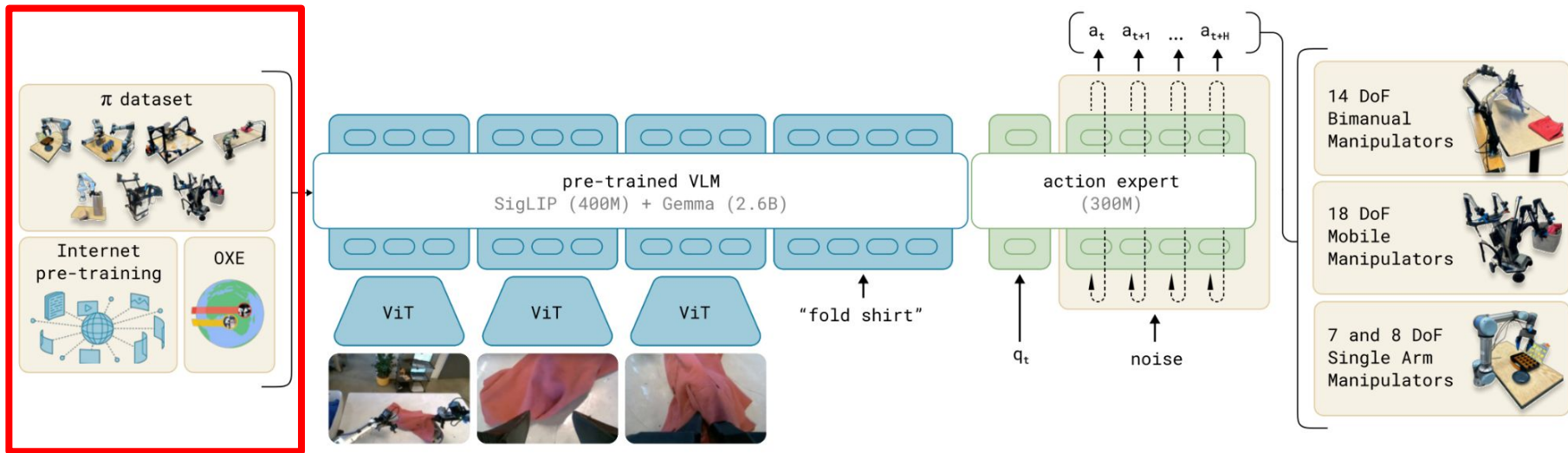


Fig. 3: **Overview of our framework.** We start with a pre-training mixture, which consists of both our own dexterous manipulation datasets and open-source data. We use this mixture to train our flow matching VLA model, which consists of a larger VLM backbone and a smaller *action expert* for processing robot states and actions. The VLM backbone weights are initialized from PaliGemma [5], providing representations learned from large-scale Internet pre-training. The resulting π_0 model can be used to control multiple robot embodiments with differing action spaces to accomplish a wide variety of tasks.

Overview

Pre-training

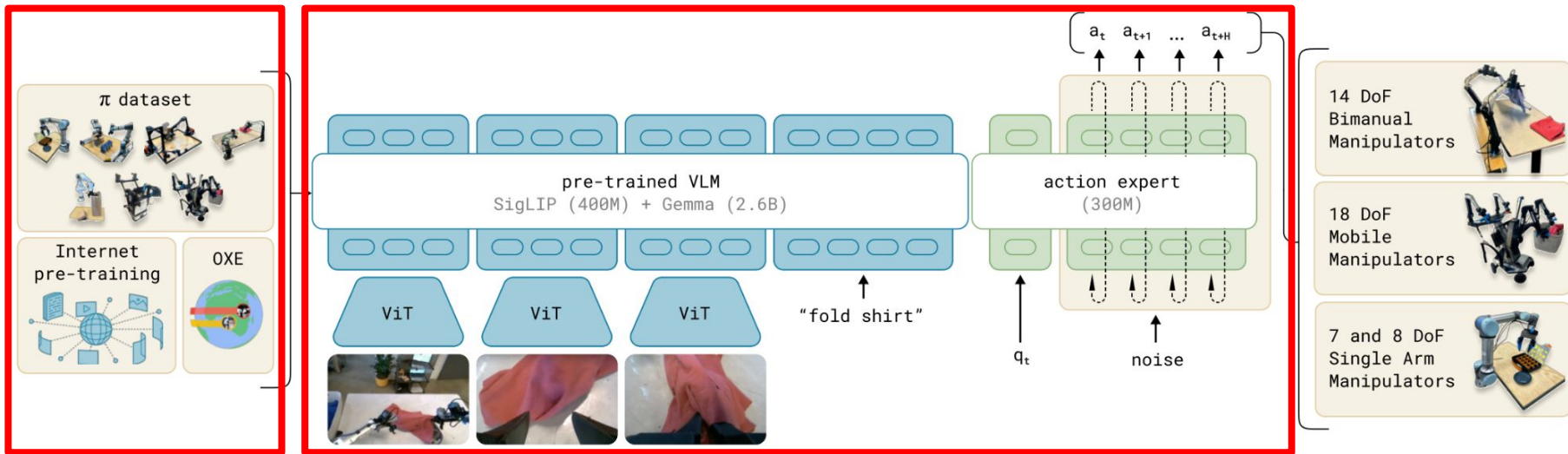


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Overview

Post-training

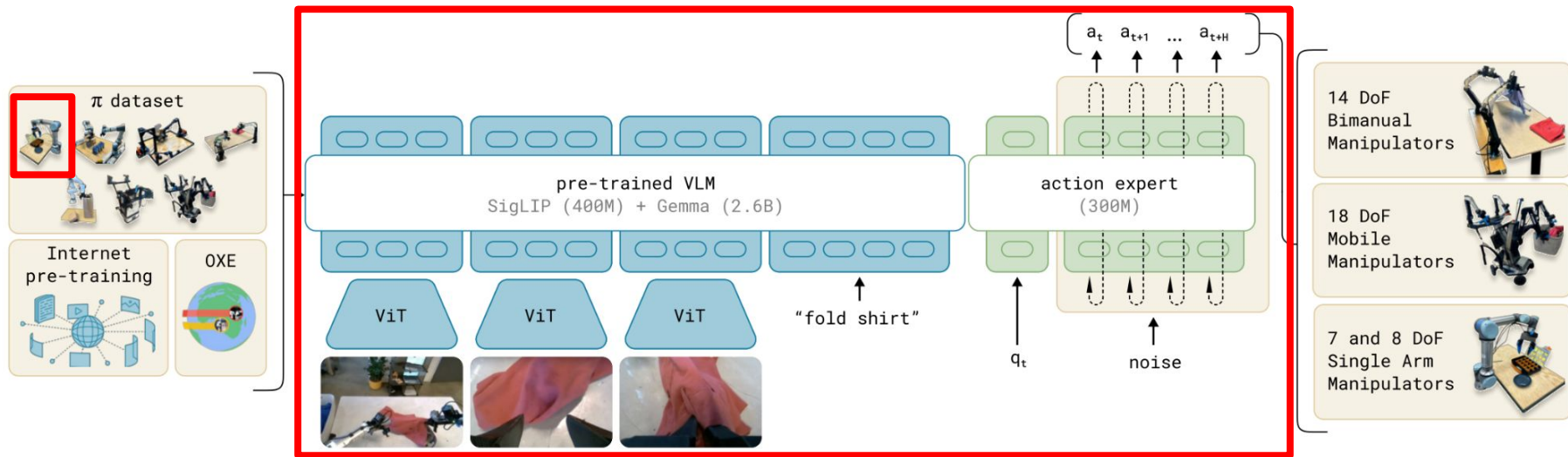


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THE π 0 MODEL

The π 0 model is primarily composed of a language model **transformer backbone**.

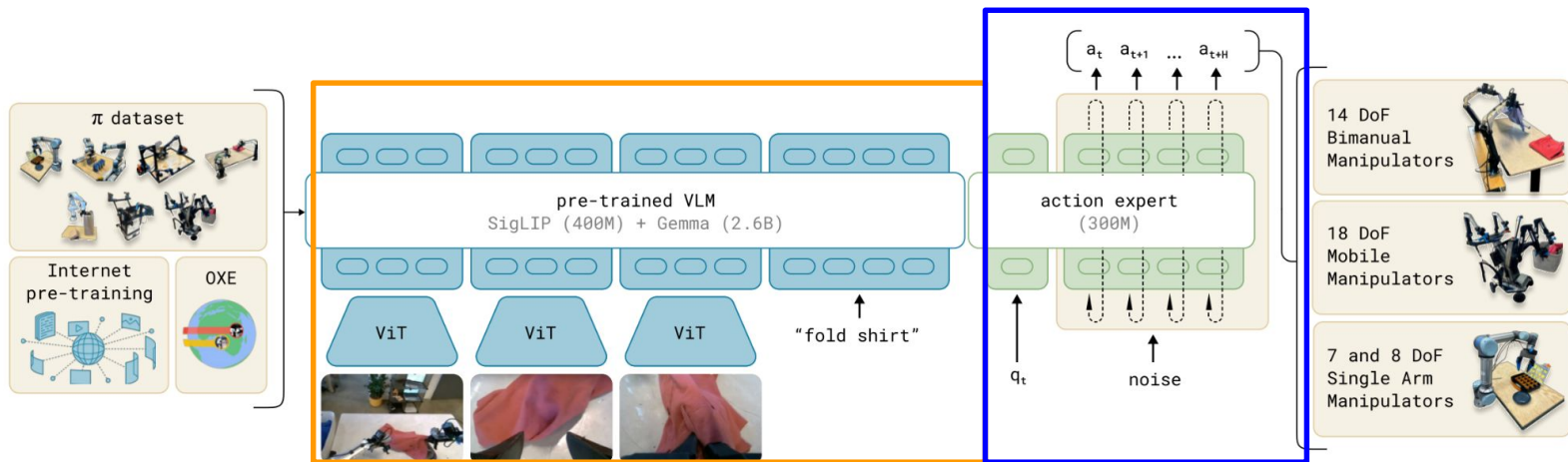
Following the standard late-fusion VLM approach [3, 11, 30], the image encoder embeds the robot's **image observations into the same embedding space as the language tokens**.

Robot-specific inputs and outputs, namely **proprioceptive state and robot actions, are added to this backbone**.

This design is similar to a **Mixture-of-Experts model** [45, 25, 12, 16] with two mixed elements:

- **The first element is used for image and text input, and**
- **The second element is used for robotics-related inputs and outputs.**

We call the second set of weights **the action expert**.



THE π_0 MODEL

This architecture is inspired by **Transfusion [59]**, which **trains a single transformer using multiple objectives**. Here, token 1 corresponds to the continuous output supervised by the flow matching loss, and token 2 corresponds to the discrete output supervised by the cross-entropy loss.

Based on the Transfusion study, we additionally found that using **a separate set of weights for robotics-related (behavior and state) tokens improves performance**.

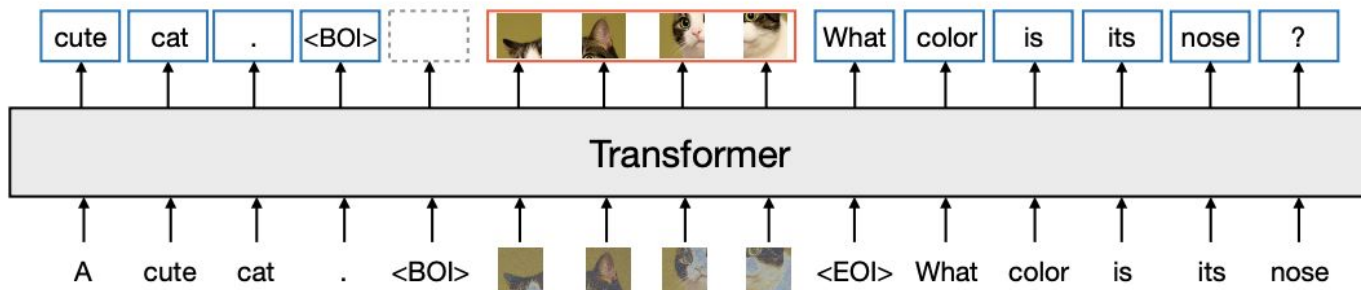


Figure 1: A high-level illustration of Transfusion. A single transformer perceives, processes, and produces data of every modality. Discrete (text) tokens are processed autoregressively and trained on the **next token prediction** objective. Continuous (image) vectors are processed together in parallel and trained on the **diffusion** objective. Marker BOI and EOI tokens separate the modalities.

THE π_0 MODEL

π_0 models the continuous distribution of actions using **conditional flow matching** [28, 32].

Flow matching provides the model with high precision and multi-modal modeling capabilities, making it particularly suitable for high-frequency, dexterous tasks.

For each action at ' in the action chunk $\mathbf{A}_t$, we have a corresponding action token that we feed through the action expert. During training, we supervise these action tokens using a **conditional flow matching loss**

Flow Matching Timesteps ($\tau \in [0, 1]$)

$$L^\tau(\theta) = \mathbb{E}_{p(\mathbf{A}_t | \mathbf{o}_t), q(\mathbf{A}_t^\tau | \mathbf{A}_t)} \|\mathbf{v}_\theta(\mathbf{A}_t^\tau, \mathbf{o}_t) - \mathbf{u}(\mathbf{A}_t^\tau | \mathbf{A}_t)\|^2,$$

Robot Timesteps

Simple Linear Gaussian probability path

$$q(\mathbf{A}_t^\tau | \mathbf{A}_t) = \mathcal{N}(\tau \mathbf{A}_t, (1 - \tau)\mathbf{I})$$

network outputs

Denoising Vector Field

$$\mathbf{u}(\mathbf{A}_t^\tau | \mathbf{A}_t) = \epsilon - \mathbf{A}_t$$

$\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

Noisy Actions

$$\mathbf{A}_t^\tau = \tau \mathbf{A}_t + (1 - \tau)\epsilon$$

THE π_0 MODEL

Non-VLM baseline model

- In addition to our main VLA model, we also trained a similar baseline model that did not use a VLM initialization for ablation experiments.
- This model, which we refer to as π_0 -small, has 470M parameters, does not use VLM initialization.
- This model is used in our comparisons to evaluate the benefits of incorporating VLM pertaining.

DATA COLLECTION AND TRAINING RECIPE

Pre-training and post-training

Each training example corresponds to a timestep, i.e., an (ot, At) tuple, so we quantify our data in terms of timesteps throughout this discussion.

- **9.1% of our training blend is comprised of open-source datasets** including OXE [10], Bridge v2 [52], and DROID [23].
- To learn more dexterous and complex tasks, **we also leverage 903 million timesteps of data extracted from our proprietary datasets.**
- In the post-training phase, **we fine-tune the model on smaller, task-specific datasets to specialize it for specific downstream applications.**

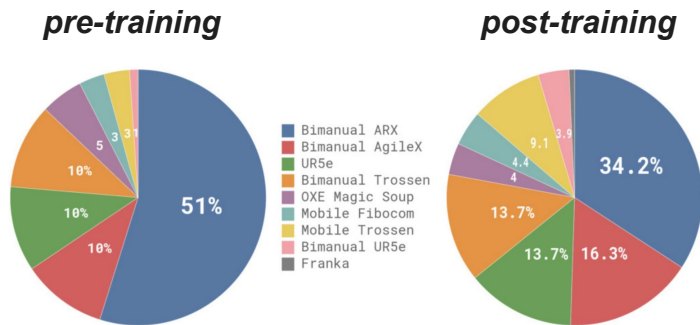


Fig. 4: **Overview of our dataset:** The pre-training mixture consists of a subset of OXE [10] and the π dataset. We use a subset of OXE, which we refer to as OXE Magic Soup [24]. The right figure illustrates the weight of the different datasets in the pre-training mixture. The left figure illustrates their relative sizes as measured by the number of steps.

DATA COLLECTION AND TRAINING RECIPE

Language and high-level policies

- More complex tasks that require semantic reasoning and high-level strategy, such as table bussing, can also benefit from a high-level policy that decomposes high-level tasks (such as “bus the table”) into more immediate subtasks (such as “pick up the napkin” or “throw the napkin into the trash”).
- Since our model is trained to process language inputs, we can use a high-level VLM to make these semantic inferences, a method that is analogous to LLM/VLM planning methods such as SayCan [2].
- We use such a high-level policy to assist our model with high-level strategy for several of our experimental tasks

DATA COLLECTION AND TRAINING RECIPE

Robot system details

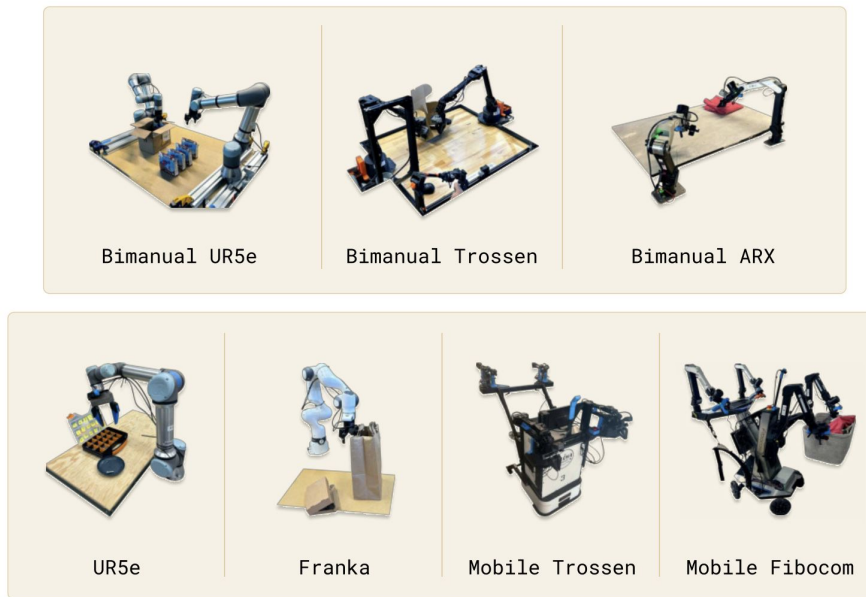


Fig. 5: The robots used in our experiments. These include single and dual-arm manipulators with 6-DoF and 7-DoF arms, as well as holonomic and nonholonomic mobile manipulators. π_0 is trained jointly on all of these platforms.

EXPERIMENTAL EVALUATION

Our experimental evaluation consists of out-of-box evaluation experiments that

1) *compare our base (pre-trained) model to alternative model designs with direct prompting*, as well as

1-1. How well does π_0 perform after pre-training on a variety of tasks that are present in the pre-training data?

1-2. How well does π_0 follow language commands?

2) *detailed fine-tuning experiments that evaluate our model on challenging downstream tasks*,

2-1. How does π_0 compare to methods that have been proposed specifically for addressing dexterous manipulation tasks?

3) *comparing it to other methods that have been proposed for dexterous manipulation*.

3-1. Can π_0 be adapted to complex, multi-stage tasks?

EXPERIMENTAL EVALUATION

1-1. How well does π_0 perform after pre-training on a variety of tasks that are present in the pre-training data?

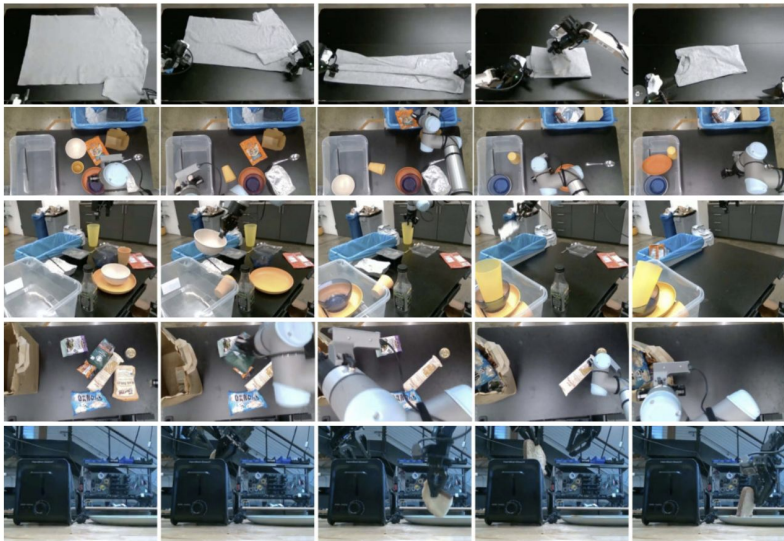


Fig. 6: **Out-of-box evaluation tasks:** To evaluate our base model, we run it after pre-training on five tasks: **shirt folding**, **bussing easy**, **bussing hard**, **grocery bagging**, and **toast out of toaster**. The tasks require a combination of dexterous manipulation, multi-stage behaviors, and semantic recognition.

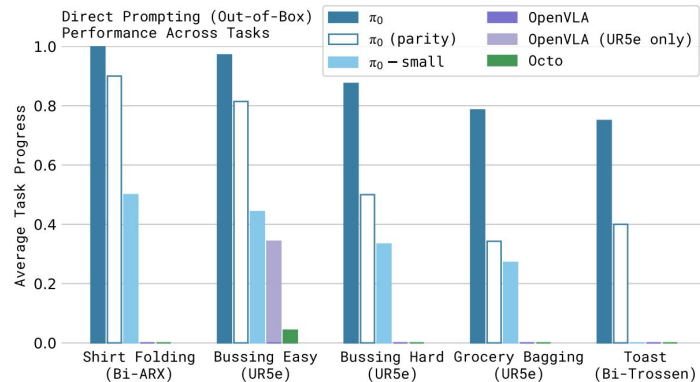


Fig. 7: **Out-of-box evaluation results:** We evaluate π_0 trained for the full 700k steps, a version trained for 160k steps that matches the number of updates for baseline models, π_0 -small, and three baselines: OpenVLA and Octo trained on all of our data, and OpenVLA trained only on the UR5e tasks (which we found to work better on UR5e tasks). Across all tasks and all comparisons, even the “parity” version of our model outperforms all baselines, and the full version of our model achieves the best results by a large margin.

EXPERIMENTAL EVALUATION

1-2. How well does π_0 follow language commands?



Fig. 8: **The tasks in our language evaluation.** We evaluate our model on 3 different language-conditioned tasks, each of which requires following a sequence of intermediate language commands. The tasks involve bussing a table (top) to put dishes in a bin and garbage in a trash bin, setting a table (middle) by taking items out of a bin, and packing a shopping bag (bottom).

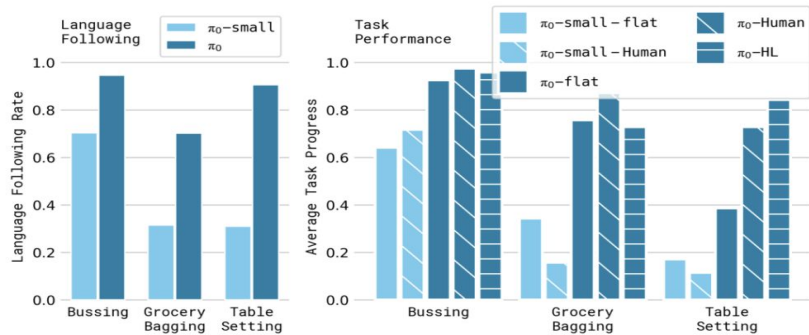


Fig. 9: **Language evaluation.** We compare “flat” versions of our policies, —flat, which receive only the overall task command (e.g., “bag the groceries”) with a method that receives intermediate commands from a human expert, —human, or a high-level VLM policy, —HL. We also compare our model to a small non-VLM variant under the “expert” condition, π_0 and π_0 -small, in terms of language following accuracy. The results show a significant improvement with π_0 from intermediate language commands provided by a human expert and to a lesser degree by an autonomous high-level policy. Notably, due to π_0 -small’s limited language following ability, overall it does not gain with the addition of a high-level expert.

EXPERIMENTAL EVALUATION

2-1. How does π_0 compare to methods that have been proposed specifically for addressing dexterous manipulation tasks?

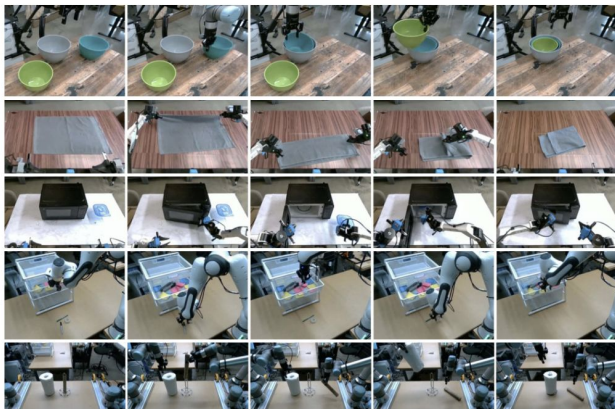


Fig. 10: **Fine-tuning evaluation tasks:** We fine-tune our model to a variety of downstream tasks that are distinct from the tasks seen in pre-training. Our tasks represent a range of similarity from the pre-training tasks, with tasks that are most similar to pre-training (**stack bowls** and **towel folding**), a task that introduces an unseen new element (a microwave), and tasks that require new motions and new object types (**Franka items in drawer** and **paper towel replacement**).

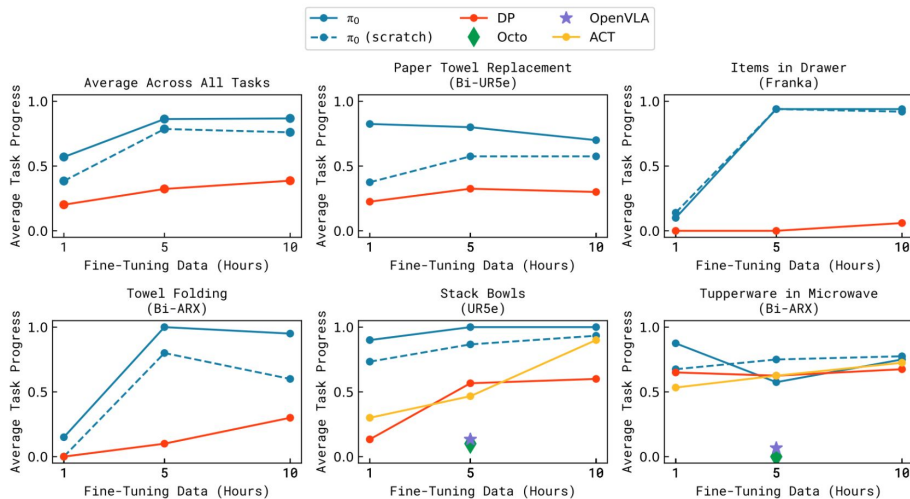


Fig. 11: **Fine-tuning with varying amounts of data.** π_0 can learn some easier tasks even with smaller amounts of data, and the pre-trained model often attains a larger improvement over the model trained from scratch.

EXPERIMENTAL EVALUATION

3-1. Can π_0 be adapted to complex, multi-stage tasks?

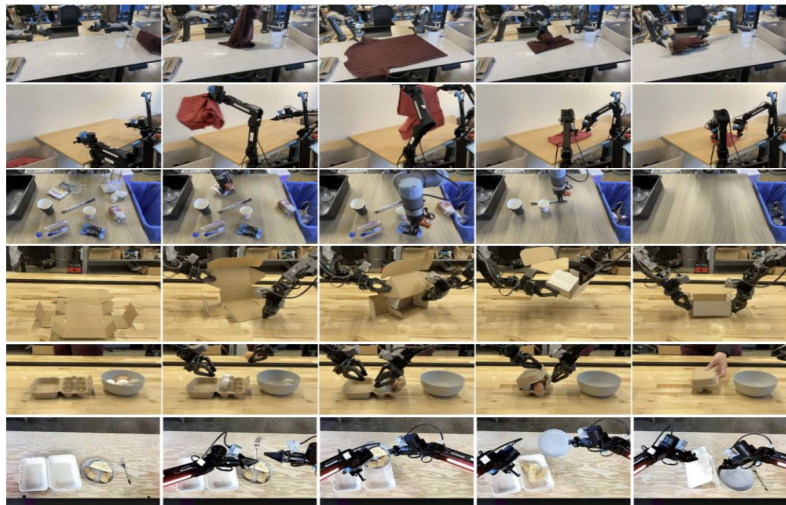


Fig. 12: We evaluate a range of complex and temporally extended tasks. This includes: folding laundry from a bin with a stationary (a) or mobile (b) robot, bussing a real lunch table (c), assembling a box (d), packing eggs into a carton (e), and packing food into a to-go box (f). These tasks require combining dozens of individual behaviors, such as grasping, stacking, folding, and flattening, generalization to a huge variety of object configurations, and complex physical properties, such as deformable objects or flexible cardboard.

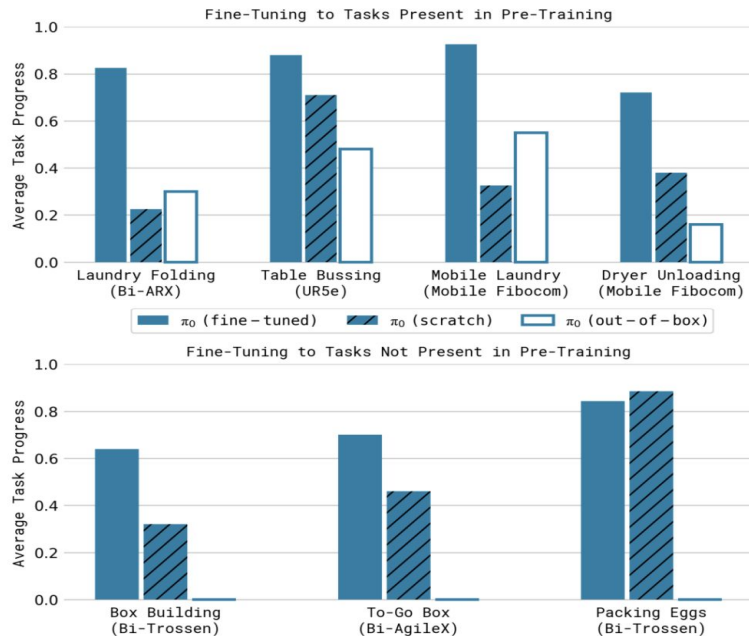


Fig. 13: **Post-training results on complex tasks** in terms of average scores over 10 trials. The full pre-trained π_0 model attains more than 50% of the maximum score across all of the tasks, and typically outperforms the ablations, with especially significant improvements on the hardest tasks.

DISCUSSION

- Our framework broadly resembles the training procedures employed for large language models, which typically consist of pre-training a base model on very large datasets scraped from the web, followed by a post-training procedure that aims to “align” the model to enable it to follow instructions and perform user commands.
- Our experiments imply that an analogous phenomenon might take place with robot foundation models, where pre-trained models have some zero-shot capabilities, but complex tasks like laundry following require fine-tuning with high-quality data.
- We hope that our results will serve as a stepping stone toward general and broadly applicable robot foundation models.

LIMITATIONS, AND FUTURE WORK

- There are a number of limitations and ample room for future work.

- 1) First, our experiments do not yet provide a comprehensive understanding of how the pre-training datasets should be composed: we combined all data available to us, but understanding what type of data is more helpful to add and how it should be weighted remains an open problem.

- 2) Not all tasks in our evaluation work reliably, and it remains unclear how to predict how much and what kind of data is needed to attain near-perfect performance.

- 3) Finally, it remains to be seen how much positive transfer there is in combining highly diverse data, particularly from different tasks and different robots.

- Although our results suggest that universal pre-trained robot foundation models might become a reality, it is left for future work to understand whether this universality extends to much more distinct domains, such as autonomous driving, navigation, and legged locomotion.